

# Advanced Statistical Learning

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## Learning From Data!

# Learning from Data - supervised learning

Given a **training set**  $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$ , learn a function

$$f : x \mapsto y$$

to predict the  $y$ 's corresponding to future  $x$ 's.

In particular, those in the **test set**  $\{(x'_1, y'_1), (x'_2, y'_2), \dots, (x'_m, y'_n)\}$ ,

# Learning from Data - supervised learning

## Outcome measurement

- ▶ Quantitative (e.g., stock price)
- ▶ Categorical (e.g., heart attack / no heart attack)

## Set of features

- ▶ Diet
- ▶ Clinical measurements

## Goal: predict outcome with set of features accurately

- ▶ Training set: observe outcome and feature for a set of objects
- ▶ Prediction model (learner): predict the outcome for new unseen objects

## Supervised: presence of the outcome variable

# Learning from Data - supervised learning

Joint distribution  $P(X, Y)$

- ▶ unknown
- ▶ Both training and testing data are generated according to  $P(X, Y)$

Hypothesis space

- ▶ Model: mapping from input to output
- ▶ Hypothesis space consists of set of models

Conditional distribution  $P(Y | X)$

Decision function  $Y = f(x)$

# Learning from Data - unsupervised learning

Only the features are observed

No measurements of the outcome

Goal: describe how the data are organized or clustered.

# Learning from Data - unsupervised learning

Training set

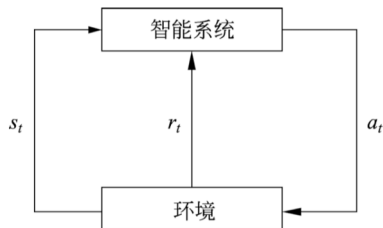
- ▶  $\{x_1, x_2, \dots, x_n\}$

Conditional distribution

- ▶  $P(z \mid x)$
- ▶ where  $z$  is latent

Decision / clustering function  $Z = g(x)$

# Learning from Data - reinforcement learning



## Five elements of reinforcement learning

- ▶  $\langle S, A, P, r, \lambda \rangle$
- ▶ state, action, transition probability, reward function, discount factor
- ▶  $P(s' | s, a) = P(s_{t+1} = s' | s_t = s, a_t = a)$ ,  
 $r(s, a) = E(r_{t+1} | s_t = s, a_t = a)$
- ▶ strategy  $\pi$ :  $a = f(s)$  or  $P(a | s)$



# Learning from Data - reinforcement learning

## Model-free

- ▶ policy based: solve for optimal strategy  $\pi^*$
- ▶ value based: solve for optimal value function
- ▶ strategy  $\pi$ :  $a = f(s)$  or  $P(a | s)$

## Model-based

- ▶ learn models for transition probability and reward function from data
- ▶ predict environmental feedback from models

# Elements of Statistical Learning - Model

Decision function

$$\blacktriangleright \mathcal{F} = \{f \mid Y = f(X)\}$$

Parameter space

$$\blacktriangleright \mathcal{F} = \{f \mid Y = f_{\theta}(X), \theta \in \mathbb{R}^n\}$$

Conditional probability

$$\blacktriangleright \mathcal{F} = \{P \mid P(Y \mid X)\}$$

Parameter space

$$\blacktriangleright \mathcal{F} = \{P \mid P_{\theta}(Y \mid X), \theta \in \mathbb{R}^n\}$$

# Elements of Statistical Learning - Strategy

## Loss function, risk function

- ▶ 0-1 loss:  $L(Y, f(X)) = 1$  if  $Y \neq f(X)$ , and  $L(Y, f(X)) = 0$  if  $Y = f(X)$
- ▶ square loss:  $L(Y, f(X)) = (Y - f(X))^2$
- ▶ absolute loss:  $L(Y, f(X)) = |Y - f(X)|$
- ▶ log loss:  $L(Y, f(X)) = -\{y \log f(x) + (1 - y) \log(1 - f(x))\}$
- ▶ multi-class log loss:  
$$L(y_1, \dots, y_K; f_1(x), \dots, f_K(x)) = -\sum_{k=1}^K y_k \log f_k(x)$$

# Elements of Statistical Learning - Strategy

Minimize empirical risk

- ▶  $\min_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i))$
- ▶ tend to overfit when sample size is small

Minimize structural risk

- ▶  $\min_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i)) + \lambda J(f)$
- ▶ introduce penalty, equivalent to regularization

# Elements of Statistical Learning - Algorithm

Should be easy if there are closed-form solution

- ▶ ordinary least squares

Often involve complicated numerical algorithms when closed-form solution does not exist

# Model assessment and selection

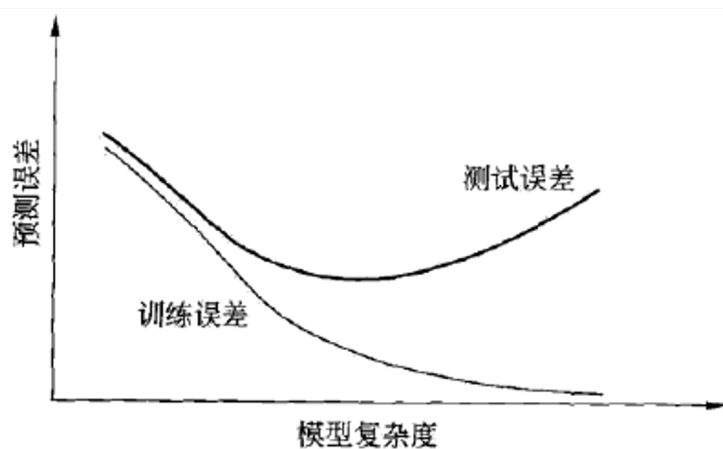
Training error vs testing error

$$\blacktriangleright R(\hat{f}) = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}(x_i))$$

$$\blacktriangleright e_{test} = \frac{1}{N'} \sum_{j=1}^{N'} L(y_j, \hat{f}(x_j))$$

When 0-1 loss is used,  $e_{test} = \frac{1}{N'} \sum_{j=1}^{N'} I(y_j \neq \hat{f}(x_j))$ , accuracy of testing set  $\frac{1}{N'} \sum_{j=1}^{N'} I(y_j = \hat{f}(x_j))$

# Model assessment and selection



# Regularization and cross validation

## Regularization

$$\blacktriangleright \min_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i)) + \lambda J(f)$$

In regression task:

$$\blacktriangleright L(\omega) = \frac{1}{N} \sum_{i=1}^N (f(x_i; \omega) - y_i)^2 + \frac{\lambda}{2} \|\omega\|^2$$

$$\blacktriangleright L(\omega) = \frac{1}{N} \sum_{i=1}^N (f(x_i; \omega) - y_i)^2 + \frac{\lambda}{2} \|\omega\|_1$$

Cross validation: training set, validation set, test set

- ▶ S-fold cross validation
- ▶ leave-one-out cross validation



# Generalizability

## Generalization error

$$\blacktriangleright R(\hat{f}) = E_p[L(Y, \hat{f}(X))] = \int_{X,Y} L(y, \hat{x}) P(X, Y) dx dy$$

## Upper bound of generalization error

- ▶ often used to measure the generalizability of learning methods
- ▶ decrease with sample size; increase with size of hypothesis space

# Generative and discriminatory approach

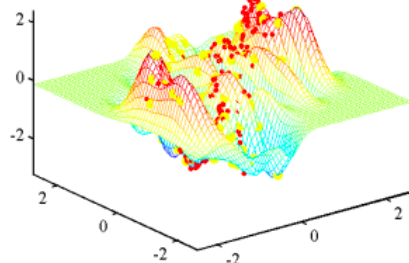
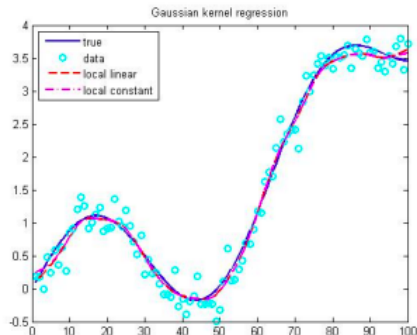
## Generative approach

- ▶ aim for  $P(X, Y)$  or  $P(X | Y)$  &  $P(Y)$
- ▶ used for data generation or classification
- ▶ Naive Bayes, Hidden Markov Model

## Discriminative approach

- ▶ aim for  $P(Y | X)$  or  $Y = f(x)$
- ▶ used for classification, prediction
- ▶ K-nearest neighbor, Perceptron, decision tree, logistic regression, support vector machine, boosting...

# Learning task - regression



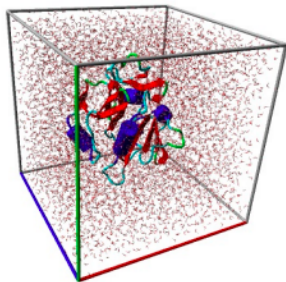
# Applications of regression



Brain computer interfaces

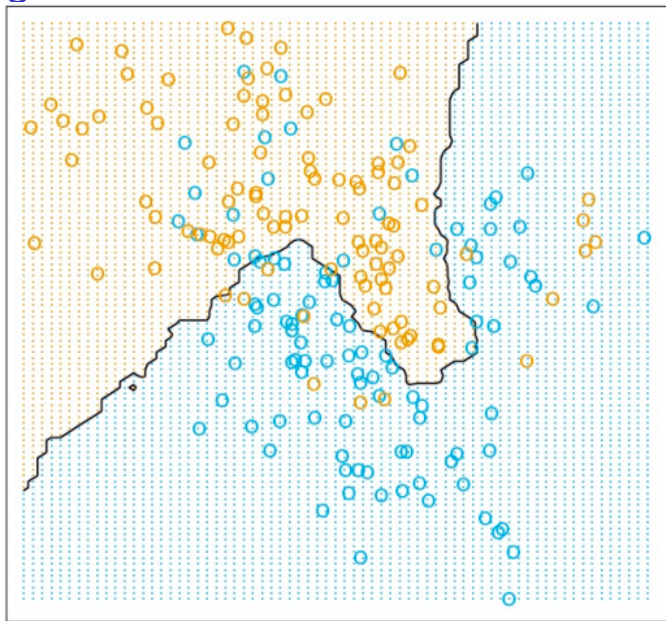


Finance



Physical modeling

# Learning task - classification



# Applications of classification



Optical character  
recognition



Face detection &  
recogniton



Autonomous vehicles



Medical diagnosis and  
monitoring

# Other learning problems - ranking



Elections

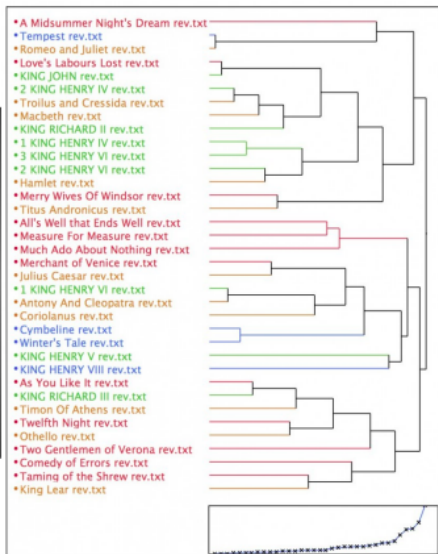
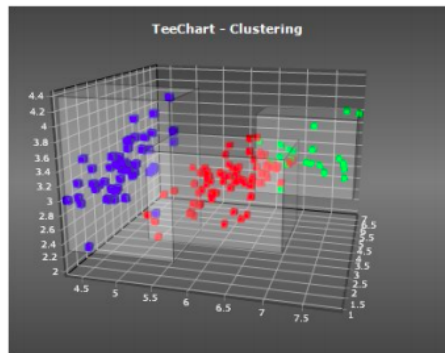


Sports



Internet search

# Other learning problems - clustering





# Case 1: Email spam

## Goal

- ▶ Predict whether the email was junk email, or “spam.”
- ▶ Design an automatic spam detector that could filter out spam

## Training Data

- ▶ 4601 email messages (examples or learning instances)
- ▶ Outcome: binary, “spam” vs “ham”
- ▶ Features: relative frequencies of 57 most commonly occurring words and punctuation marks in the email message

# Examples of Spam Email

I searched my gmail spam folder, and here are some of the common spam email titles...

- ▶ **!!URGENT!!**
- ▶ Congratulations you have been confirmed!
- ▶ Urgent message for you
- ▶ Free \$5000 for you!
- ▶ Personal loan up to \$15000

# Case 1: Email spam

Average percentage of words or characters in an email message

|       | george | you  | your | hp   | free | hpl  | !    | our  | re   | edu  | remove |
|-------|--------|------|------|------|------|------|------|------|------|------|--------|
| spam  | 0.00   | 2.26 | 1.38 | 0.02 | 0.52 | 0.01 | 0.51 | 0.51 | 0.13 | 0.01 | 0.28   |
| email | 1.27   | 1.27 | 0.44 | 0.90 | 0.07 | 0.43 | 0.11 | 0.18 | 0.42 | 0.29 | 0.01   |

Our learning method has to decide which features to use and how:

- ▶ if ( $\% \text{george} < 0.6$ ) & ( $\% \text{you} > 1.5$ ) then spam, else ham
- ▶ if ( $0.2 * \% \text{you} - 0.3 * \% \text{george} > 0$ ) then spam, else ham
- ▶ .....

Cost of errors

- ▶ filtering out ham as spam - ohh, no!
- ▶ recognize spam as ham - not desirable, but OK

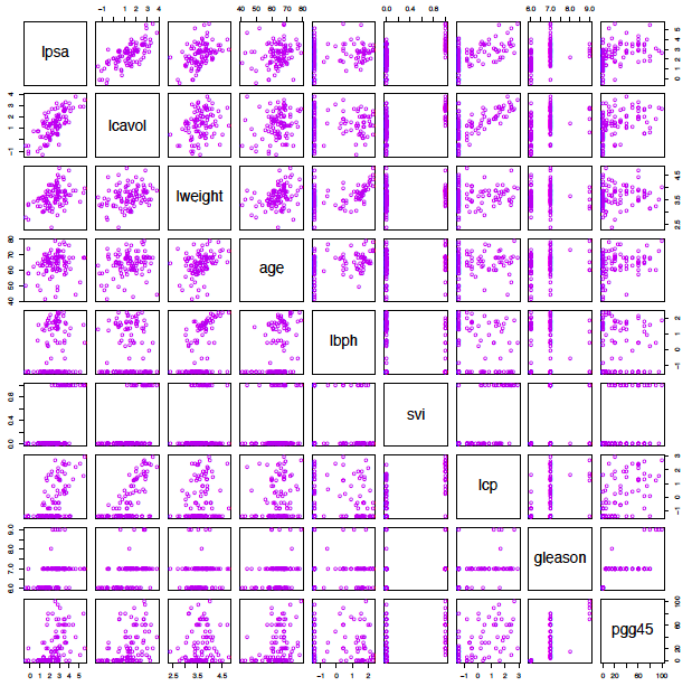
## Case 2: Prostate Cancer

### Goal

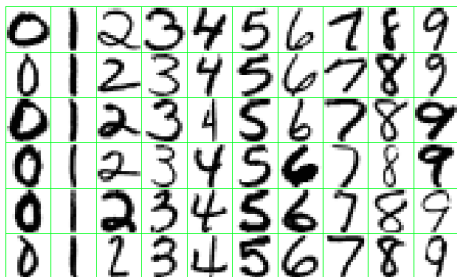
- ▶ predict the lpsa from a number of measurements: lcavol, lweight, age, lbph, svi, lcp, gleason, pgg45

### Training Data

- ▶ 97 male who were about to receive a radical prostatectomy.
- ▶ Outcome: continuous, log of prostate specific antigen (psa)
- ▶ Features: lcavol, lweight, age, lbph, svi, lcp, gleason, pgg45



## Case 3: Handwritten Digit Recognition



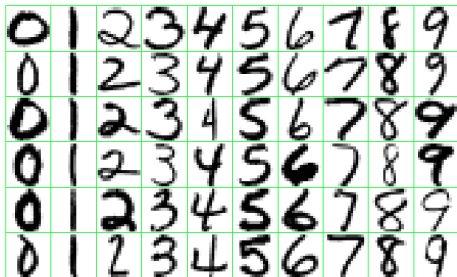
### Goal

- Predict the identity of each image (0, 1,..., 9) quickly and accurately.

### Training Data

- Images isolated from postal envelopes
- Outcome: categorical, 0-9 image identity
- Features:  $16 \times 16$  matrix of pixel intensities (from 0 to 255)

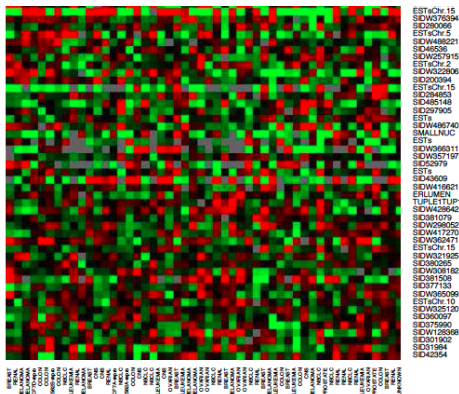
## Case 3: Handwritten Digit Recognition



Classification problem:

- ▶ Error rate needs to be kept very low to avoid misdirection of mails
- ▶ Some objects can be assigned to a “don’t know” category, and sorted instead by hand.

# Case 4: DNA Expression Microarrays



Unsupervised learning problem

- ▶ Cluster together genes or samples

Training Data

- ▶ Each row is a gene, each column is a tissue sample
- ▶ Level of expression: green - negative; red - positive



# Plans for Next Class

Reading (optional): The Elements of Statistical Learning, Chapter 2, Overview of Supervised Learning, p9-p38

Think about potential book chapter to present